

Privacy-Preserving Federated Approach for Primary Pancreatic Malignancy Detection

ABSTRACT

Pancreatic cancer is one of the deadliest malignancies worldwide, where timely detection can significantly improve patient survival. But building effective AI-based diagnostic models requires large and diverse clinical data from different hospitals, which cannot be centrally aggregated due to strict privacy policies (e.g., HIPAA, GDPR). In this study, we present a privacy-preserving federated learning (FL) framework that trains a unified deep neural network model for primary pancreatic malignancy detection using structured clinical records from the SEER program. Here, each simulated hospital node trains the model by keeping its own data locally; only the model parameters are aggregated via a Federated Averaging (FedAvg) algorithm on a central server, thus eliminating the need to share raw patient data at any stage. Data preprocessing (missing-value handling, encoding, and scaling), feature selection, class imbalance handling, and a relatively lightweight ANN-based architecture ensure the effectiveness and stability of this framework. Experimental comparisons with centralized training methods shows that the federated model is able to maintain nearly equivalent accuracy despite data heterogeneity across nodes—indicating the realistic potential of multicenter healthcare collaboration. This work demonstrates that it is possible to build robust AI-based cancer detection models while fully preserving patient data privacy. Beyond pancreatic cancer, this framework is applicable with other sensitive medical-based prediction problems, which could accelerate the adoption of safe, scalable, and policy-compliant AI in healthcare.

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Chapter 1

Introduction

This study propose a privacy-preserving federated learning framework based on SEER data to rapidly and accurately detect primary pancreatic malignancy while preserving patients privacy.

1.1 Introduction

Pancreatic cancer is a very dangerous disease, and many people finds out about it when it is already to late. When this hapen, treatment do not work very wel, and people do not live for a lonng time. Finding pancreatic cancer early is very impportant because it help doctors choose the right treetment and making better decisions. This project trys to find the main pancreatic cancer early so pacients could get beter care and has a beter chance to live loger.

Hospital have a lot of patient infformation, but they can not share it because of priveacy rules. So, the data staying in many diferent hospitals. Federeted learnning help solve this problem by leting hospitals work together without sharing patient details. In this study, SEER cancer data is use with federeted learning to finds pancreatic cancer. This mathod keeps patient info safe and still give good and acurate results.

1.2 Motivation

This research is mainly about keep patient infformation safe and private. Hospitals has very important and sensitive data about patiants, and sharing this data can risky. Old systems asks hospitals to send all their data to one place, which can causes problems like data leakage. In this study, hospitals keeps the data with them and ddoes not send it anywhere. They only share llearning updates, which helping protect privacy and folow safety rules. This makes doctors and hospitals feling more secure when using new tecnology.

Pancreatic cancer is very dengerous because it grow fast and is offten found to late. When the cancer is found early, doctors can treats it beter and helps patients live longer and healter lives. This resarch try to find pancreatic cancer early by using information from many hospitals at the same time. It also help doctors understand the diseese better, so they can chose the best tratment for each patient.

This study also help hospitals work together in a safe way. Each hospital help train the same model while keping it own data private. When many hospitals joins, the model learns from diferent kinds of patients and become stronger and smarter. This teamwork help build beter cancer detecscion system that can be using in real hospitals in the future.

1.3 Objectives

The purpose of this research is to reaching specific objectives that will be make the project worthwhile. The key objectives are:

1. The main goal of this research is to protects patient privacy by keeping medical data safe and secure while using it for learning. This is done by allowed hospitals to keep data locally and not sharing raw patient information.
2. The second goal of this study is to detecting pancreatic cancer at an early stage by malignancy detection so that doctors can plan beter treatment and improve patient survival.
3. Another objective is to building a decentralized learning system where multiple hospital can train the same model without sending data to a central server.
4. Finally, this research aim to improve collaboration among hospitals so that the model become more accurate and useful in real-world cancer detection.

1.4 Methodology

In this study, we made a safe and smert system to find early pancreatic cancer using SEER data. The data was diviide into diferent hospital groups so it look like real hospitals. Each hospitel kept it own pasient data private and did not shares it with others. All the data was cleand, fixed, and prepard in the same way so the learnning was fair and easy to understand. A deep leearning model was use to help find early panncreatic cancer. The hospitals leern together using federeted learning, where only the leaning results were shared, not the patient data. A main server colected these results and made the model beter for everyone. This system kep patient infomation safe, folow privacy ruls, and still give good results. It also help hospitels work togather and can be used in real heelticare systems in the future.

1.5 Project Outcome

This project show that federeted learning can be used to find paencreatic cancer in a smart and safe way. It use SEER cllinical data to help docttors find cancer for the firrst time, understtand tumor signs, and studiies patient details like age and tumor type. The system do not coilect all data in one place. Instead, each hospital keep its own data safe, and no patient infomation is sharing. This make the system private and secure while still helping doctors learn from the data.

Many tests was done usiing data from diferent hospitals, and the results was very good. The model could found cancer correctly, predict important clinical signs, and study risk using patient and tumor information. Federated learning help hospitals work together without sharing private data. The results was almost as good as using all data in one place, but much safer. This project show that federated learning is very useful for healthcare and can also be use in the future for other diseases, helping patients while keeping ther information safe.

1.6 Organization of the Report

Chapter 1 (Introduction) present the overall overview of the project by explaining the background and importnce of pancreatic cancer, the motivation for using privacy-

preserving federated learning in healthcare, the problem statement, research objectives, scope of the work, and a brief outline of the report structure.

Chapter 2 (Background) provide the necessary theoretical and practical foundation for the study. It review existing research on pancreatic cancer detection, machine-learning-based medical diagnosis, and federated learning, and discuss the limitations of centralized approaches. A gap analysis is include to justify the need for privacy-sensitive and collaborative medical data analysis.

Chapter 3 (Research Methodology) describe the research process in detail, including dataset selection, data preprocessing, feature selection, and model development. It explain the federated learning framework, neural network design, local training, parameter sharing, FedAvg agregation, and the privacy asurance mechanisms used in the system.

Chapter 4 (Implementation and Results) cover the system implmentation, development environment, experimeental setup, and evaluauting procedures. It present the results of primary maliignancy detection and cllinical feature-based analysis, alongs with a comparative study between federeted and centrallized models for highllight performance, accuracy, efficiency, and privacy benefits.

Chapter 5 (Engineering Standards and Design Challenges) discuss compliance with international standaards such as ISO, IEEE, HIPAA, and GDPR. It also adress ethical concerns, patient privacy, system reliabillity, sustainibilty, and key design challenges of federated learning, inclding non-IID data, communication overhead, scalability, and regulatory complliance.

Chapter 6 (Conclusion) sumarize the key findiings of the reseearch, confirms the efectivness of federated learning for privacy-presrving pancreatic cancer detection, note the limitaations of the current work, and outline future research directtions such as real-world hospittal deployment, enhancced privacy teknikues, personalised federated learning, and multiimodal data intiegation.

Chapter 2

Background

The literature review highlights federated learning's potential for privacy-preserving multi-institutional cancer diagnosis whilst noting challenges in data heterogeneity, computational costs, model personalization, and clinical interpretability based on some related papers.

2.1 Introduction

Pancreatic cancer is a very dangerous disease, and it is often found too late. Many people do not feel clear symptoms at the beginning, so doctors can not find it early. When the cancer is found late, treatment does not work well. Because of this, scientists want to use smart computer systems to help find the disease early. These systems need a lot of patient information to learn properly. But hospitals can not share patient data easily because of strict privacy rules. This makes it hard to build one big system using all the data. To solve this problem, a new idea called federated learning is used. With this method, many hospitals can work together to train a model, but they keep the patient data safe inside their own hospitals.

Many studies show that computer systems can help find pancreatic cancer early. Some systems use scans and medical reports and work almost as well as doctors. But most of them need all the data in one place, which is not safe. Federated Learning helps because hospitals do not need to share raw data. Research also shows that this method works well and keeps privacy protected. Using large cancer data like SEER is very helpful too. Overall, Federated Learning is a good and safe way to help doctors find pancreatic cancer early and protect patient information.

2.2 Literature Review

Youngjung Kim [1] published a comprehensive review on the clinical utility of artificial intelligence (AI) in early detection and disease progression assessment of pancreatic cancer (PDAC) in 2025. The study highlights the progress in early disease detection using deep learning, computed tomography (CT) imaging, radiomics, and blood-based biomarkers (such as microRNAs or miRNAs and circulating tumor DNA or ctDNA). The AI-based radiomics model was able to detect disease approximately 398 days before clinical diagnosis, achieved an AUC of 0.99 for miRNA-based assays, and had a significantly shorter disease-free survival time ($HR = 5.45$) in ctDNA-positive patients. The study also notes that federated learning and privacy-preserving frameworks can play an important role in enabling collaboration across multiple institutions while preserving confidentiality. However, data standardization, model generalization, and regulatory approval remain major challenges.

Alexander Chowdhury et al. [2] conducted a comprehensive review of the application of federated learning (FL) in oncology and cancer research in 2022. They analyze the FL

algorithms, frameworks, and privacy protection methods use in various cancer datasets such as BraTS, ISIC, TCGA, and Wisconsin Breast Cancer. The study finds that FL provides opportunities for collaborative research without sharing sensitive patient information, addressing data heterogeneity and privacy issues. Although high performance have been achieved in various cancer detection and segmentation tasks, most study are still limited to the early experimental stage. Lack of data labeling, domain shift, and lack of clinical trials are identified as major challenges. Scalability, interoperability, and robust privacy frameworks are emphasized for effective application of FL in the future.

Hongyi Pan et al. [3] develops an AI-based framework called Cyst-X in 2025, which predict pancreatic cancer risk using multi-center MRI data. This framework uses data from patients with IPMN (intraductal papillary mucinous neoplasms)—1,461 MRI scans (T1W and T2W) from 764 patients across 7 institutions. It use PanSegNet for pancreatic segmentation and DenseNet-121 for cancer risk classification. The central model have an AUC of 0.82, which is better than the Kyoto guidelines (AUC 0.75) and experienced radiologists. Even using federated learning (FedAvg and FedProx), the AUC remain almost the same (≈ 0.81), and data privacy is also intact. This is a practical example of collaborative AI model training across multiple institutions while maintaining confidentiality. However, high computational complexity, data heterogeneity, and the need for clinical validation remains.

Nuha Alruwais et al. [4] developed a federated learning framework base on the Gray Wolf Optimization (GWO) algorithm in 2025, which is use for pancreatic tumor classification in the Internet of Medical Things (IoMT). The method achieve 98.9% accuracy on a CNN model trained on 500 CT images, outperforming the conventional FedAvg and Genetic CFL methods while also reducing communication costs. Although this method can maintain privacy and data balance, high computational cost and hyperparameter sensitivity remains significant limitations.

Ziliang Hong et al. [5] proposed a federated learning framework for the first time using MRI data for segmentation of the head, body, and tail of the pancreas. They evaluates the performance of the U-Net, Attention U-Net, and Swin-UNETR models using 711 T1W and 726 T2W scans from 7 medical institutions. The Attention U-Net model with the FedAvg algorithm shows the highest performance—a Dice score of 0.67 on T1W data. Although the model performed well on a diverse set of data from different institutions, high computational demands and performance variations across different parts of the pancreas were identified as challenges.

Zezhong Ma et al. [6] proposed a federated learning-based assisted diagnosis model in 2022, aiming to accurately predicts the time and location of recurrence in cancer patients while protecting patient privacy. In this model, a convolutional neural network (CNN) is incorporate into a federated learning framework, which allow multiple hospitals to train the model collaboratively without sharing raw data. The model, build using clinical data from multiple hospitals and six important health indicators (such as tumor, immune, and nutritional scores), achieves an accuracy of over 90%, which is superior to conventional

machine learning methods. Although this methods able to increase the accuracy of diagnosis while maintaining privacy, it limitations include dependence on data quality, high computational demands, and scalability issues dues to the limited number of participating institutions.

Jinshou Yang et al. [7] publishe an impotent review on early detection and screening stratgies for panccreatic cancer in 2021. The study highlight recent advances in biomarkers, imaging, and liquid biopsy technology. It states that the prognosis of pancreatic cancer is generally poor due to late detection, so effective early screening for high-risk populations is essential. In addition to improvements in endoscopic ultrasound (EUS), MRI, and serum markers such as CA19-9, new biomarkers, ctDNA, miRNA, and exosome-based tests are also mentioned. Artificial intelligence, metabolomics, and nanotechnology are identified as potential future tools. However, no single biomarker has yet demonstrated sufficient sensitivity and accuracy, so further clinical validation and the development of cost-effective screening methods are needed.

Elisabetta Grazia Tomarchio et al. [8] developed an AI-enhanced in silico workflow in 2025 that can rapidly identify inhibitors of the KRAS G12D mutation — a mutation commonly found in pancreatic, lung, and colorectal cancers. The study used a combination of pharmacophore filtering, GNINA deep learning molecular docking, and molecular dynamics simulations to screen FDA-approved and commercial compounds. Three compounds (ChEMBL4968507, Z4499936471, and MCULE-3539542683) demonstrated high binding affinity and stability comparable to the MRTX1133 inhibitor. This approach opens up the possibility of rapid and effective drug repurposing. However, as the study is entirely computational, the need for experimental validation remains a significant limitation.

Chun-Ju Yang et al. [9] developed an integrated photonic chip-based optical neural network called MediONN in 2024, which is used for semantic segmentation to quickly and accurately detect pancreatic lesions. The model uses a silicon photonic microring resonator array to perform matrix multiplication quickly and is trained on clinical image data using a U-Net architecture. The MediONN model achieves 97.98% accuracy, 95.23% sensitivity, and 99.2% specificity, provides results in just 0.37 milliseconds, and consumes much less power than conventional electronic systems. Its limitations include the complexity of chip fabrication, scalability challenges, and the need for extensive validation on data from multiple centers.

Qiuxia Yang et al. [10] developed a non-invasive, multicenter imaging–AI pipeline in 2025 that can quantify tumor fibrosis and predict the likelihood of response to chemotherapy in patients with pancreatic ductal adenocarcinoma (PDAC). Tissue segmentation was performed using H&E WSI data from 361 resectable cases from TCGA, SYSUCC, and XYCSU and validated using RNA-seq data (n=51). A CT radiomics model (SVM, 15 selected features) was then trained and tested on the XYCSU data. Patients with high fibrosis had a higher survival rate; the external AUC of the CT model was ≈ 0.718 , and both progression-free and overall survival were significantly higher in the high fibrosis group than in patients receiving AG chemotherapy. The strengths of this study are the

multicenter validation, histopathology, and transcriptomic evidence, and interpretable radiomics features. However, its limitations include retrospective design, limited sample size, use of only one slide for each patient, and the need for prospective validation before clinical application.

Simon Tongbram et al. [11] developed a hybrid deep learning framework called PANDA-Net in 2025, which significantly improved the accuracy and speed of pancreatic cancer detection and segmentation using computed tomography (CT) and magnetic resonance imaging (MRI) images. The model uses Adaptive Contrast Enhancement with Residual Learning (ACERL) for fine tissue structure retrieval, Graph Attention-based CNN (GACNN) for spatial relationship understanding in segmentation, Swin Transformer with Adaptive Token Pruning (Swin-ATP) for efficient classification, and Sand Cat Optimization (SCO) for dynamic hyperparameter tuning. PANDA-Net achieves 99.55% Dice score and 99.6% classification accuracy when tested on CT, T1-, and T2-weighted MRI datasets with a five-way cross-validation method, which is higher than U-Net, DeepLabV3, ResUNet, and Attention U-Net. The main strengths of the study are multimodal validation, the combination of transformer and attention architectures, and computational efficiency. However, limitations include reliance on retrospective data, limited diversity of datasets, and lack of testing in a real clinical setting. Therefore, future multicenter and prospective validation is essential.

Po-Ting Chen et al. [12] conducted and validated a deep learning system in 2023 for pancreatic cancer detection in CT images in a national population-based study. A 3D CNN model was trained using 3,208 abdominal CT scans (including cancer and non-cancer cases) from the Korean National Health Insurance Service database. The model achieved an AUC = 0.989, sensitivity 94.9%, and specificity 97.5% in external validation, and outperformed radiologists in detecting small tumors at an early stage. This system may play a major role in screening and clinical decision support at the public health level. However, limitations include a lack of model interpretability, domain shift due to differences between CT scanner vendors, and underrepresentation of rare tumor types, which indicate the need for future multicenter validation.

Hisham Daher et al. [13] (2024) published an important review of AI-enabled strategies for early detection of pancreatic cancer. They discussed the use of biomarkers (CA19-9, CEA, CA242, miRNA, cfDNA, extracellular vesicles), imaging modalities (CT, MRI, EUS, PET), and various machine and deep learning approaches (CNN, DLAD, CAD, ensemble models). For example, a multi-analyte panel from 2020 achieved 92% accuracy, while an EUS-based deep learning CAD approach achieved approximately 94%/92% test and validation accuracy. However, practical applications remain limited—such as data scarcity, imbalance, radiomics reproducibility issues, model interpretability and bias, and data privacy and scalability.

Cao K. et al. [14] (2023) developed a system called PANDA (Pancreatic Cancer Detection with AI), which is capable of detecting early pancreatic lesions using non-contrast CT images. After training with data from 3,208 patients and validating with data from 6,239 patients across 10 centers, PANDA achieved an AUC of 0.986–0.996, a sensitivity of

92.9%, and a specificity of 99.9%. It demonstrate 34.1% higher sensitivity than radiologists and showed equivalent performance to contrast-enhanced CT. The system may been able to detect many undiagnosed cases if it is incorporated into clinical workflows. However, the lack of international diversity in the training data, low performance in rare lesions, and the need for prospective validation has been identified as major limitations.

Placido D. et al. [15] (2023) developed a deep learning algorithm called CancerRiskNet, which predicts pancreatic cancer risk by analyzing a patient's long-term disease history. They used data from 8.1 million Danish patients (23,985 cases) and 3 million US veterans (3,869 cases). The Transformer and GRU models achieved AUROC = 0.879 (Transformer) in disease prediction at 36 months, which is better than Bag-of-Words (0.807) and MLP (0.845). The AUROC was 0.71 when directly applied to the US data, but after retraining, it increased to 0.775. The most important prognostic indicators were jaundice, biliary tract disease, abdominal pain, and type 2 diabetes. This framework can help in early surveillance of high-risk patients. However, differences in healthcare systems, low accuracy in long-term predictions, and the need for federated or locally adapted models have been identified as major challenges.

Kenner B. et al. [16] (2021) published a comprehensive summary and expert opinion on the role of artificial intelligence (AI) in early detection of pancreatic cancer. It was prepared for the 2020 “AI and Early Detection of Pancreatic Cancer Virtual Summit”. The report highlights progress and challenges in five key areas: (A) imaging and genomic/biomarker features for early detection, (B) AI and machine learning approaches, (C) ongoing research initiatives, (D) organization and collaborative models, and (E) future directions. It states that the main reason for the poor prognosis of PDAC is late diagnosis. Therefore, risk assessment, screening of high-risk groups, and the integrated use of biomarkers (e.g, CA19-9), imaging (EUS, MRI), and modern AI tools are essential for early detection. AI-assisted radiomics, disease kinetic analysis, and discovery of new biomarkers are creatin significant opportunities. However, lack of data, lack of standardization, model bias, and complexity of integration into clinical applications remain major obstacles. The authors call for acelerating the application of AI in real-world medical settings through multi-facted and data-driven collaboration.

Mammen P. M. [17] (2021) presented a comprhensive survey on federated learning (FL), which explains in detail its structure, types, applications, challenges, and new research directions. The study cover the main steepes of federeted learning—client sellection, parametter propagation, local model training, and global inteigration. The types of FL are liste as vertical, horizoontal, transfer, crossilo, and cross-device models. Important appplication areas include heealthcare (EHR-based models), transporatation, economicas, and languagse processing, where colaborative model development are possible without sharing data. The challenges includes comunication complexity and data heterogeneiy, while security threats include data lakage, model poisoning, and backdoor attaacks. To address these, secure commputation, differenttial privacy, and particiipant-level filterring has been proposed. Recent advances includes “one-shot FL,” inccentive mechanisms, “FL as a Seervice,” asynchrounous FL, and blockccchain integration. Although FL more efective

in protecting privacy than centralized models, robust defense mechanisms, efficiency, and interoperability remain major challenges.

Kairouz P. et al. [18] (2021) published an important review on federated learning, which is considered a fundamental reference in the field. This study discusses the basic principles, architecture, challenges, and future research directions of FL. They divide FL into two main categories—cross-device and cross-silo. Key topics discussed include efficiency (non-IID data handling, communication compression, personalisation), privacy (differential privacy, secure aggregation), security (attack prevention), fairness (bias reduction), and system design (scalability, on-device implementation). Also, privacy-preserving approaches based on decentralized and split learning, incentive mechanisms, and secure multiparty computation are highlighted. Key open research problems include robust aggregation, efficient training on heterogeneous data, maintaining fairness, combining FL with other privacy technologies, and ensuring large-scale practical implementation.

Alsaman H. et al. [19] (2024) proposed a federated learning framework for breast cancer detection, where a model is trained collaboratively across multiple institutions using deep convolutional neural networks (DCNN). The datasets used included VinDr-Mammo, CMMD, and INBreast. The model achieved 98.9% accuracy, 95.0% sensitivity, and 98.0% specificity in a five-client federated setup. The core components of the model were FedAvg aggregation, image preprocessing (CLAHE, background removal, feature extraction, and fusion), LeNet-based CNN architecture, and Grad-CAM interpretability. Homomorphic encryption was added to data security, which has a minor impact on performance. However, high energy consumption (~120 kWh per node) and the need for further efficiency improvements were noted in practical applications.

Ali M. S. et al. [20] (2024) conducted a literature review on federated learning in healthcare. They analyzed over 600 studies and presented a summary of 100 articles. They categorized the architectures of FL (horizontal, vertical, transfer, domain adaptation, multitask, meta-learning) and explained their applications in disease detection, EHR analysis, medical imaging, and wearable device-based health monitoring. FL protects data privacy, enables multi-party collaboration, and enhances the generalization ability of models to diverse data. However, non-IID data handling, communication, and computational efficiency, bias reduction, model interpretability, and scalability remain major challenges. They also mentioned issues related to data leakage, model poisoning, model theft, and fairness. Differential privacy, secure aggregation, and robust aggregation mechanisms are proposed as preventive strategies. For the future, the authors emphasize standardization, a balance between privacy and fairness, and the development of methods that can be used with low resources.

Tang and Wan [21] developed a prognostic model in 2025 for patients with papillary renal cell carcinoma (pRCC) using clinical data from the SEER database. They analyzed data from 1,788 patients and built a nomogram based on factors like age, sex, surgical method, pathological grade, and TNM stage. The model showed good predictive power, with AUC values of 0.798, 0.781, and 0.754 for 1-, 3-, and 5-year cancer-specific survival in the training set. Validation AUCs were slightly lower but consistent. This study demonstrates

the potential of SEER-based nonograms to aid clinical decision-making in pRCC prognosis following surgery.

Lin-Feng et al. [22] conducted a population-based retrospective study in 2025 on primary malignant conjunctival tumors (PMCT) using SEER data from 1975–2018. Analysing 2,853 cases, they found lymphoma (39.6%) to be the most prevalent subtype, followed by squamous cell carcinoma and melanoma. The study reported a peak PMCT incidence rate of 0.136 per 100,000, with increased risk in patients aged ≥ 75 . Multivariate Cox regression identified age, SEER stage, histology, and surgery as significant predictors of survival. Surgery and radiotherapy were associated with better outcomes, especially for localised cases. The findings provide valuable epidemiological insight and may inform early diagnosis and treatment strategies.

Table 2.1: Literature Review Summary

Author(s)	Year	Title	Methodology	Key Findings
Youngjung Kim [1]	2025	Clinical Utility of Artificial Intelligence in Early Detection and Disease Progression Assessment of Pancreatic Cancer	Comprehensive review of AI-based methods including CT imaging, radiomics, miRNA and ctDNA biomarkers, and deep learning models for early PDAC detection	AI-based radiomics detected PDAC up to 398 days before diagnosis; miRNA assays achieved AUC of 0.99; ctDNA-positive patients showed significantly worse survival; federated learning highlighted as privacy-preserving solution
Alexander Chowdhury et al. [2]	2022	Federated Learning in Oncology and Cancer Research: A Review	Review of federated learning algorithms, frameworks, and privacy methods applied to datasets such as BraTS, ISIC, TCGA, and WBC	FL enables collaborative cancer research without data sharing; strong performance in detection and segmentation tasks, but mostly limited to experimental studies
Hongyi Pan et al. [3]	2025	Cyst-X: AI-Based Pancreatic Cancer Risk Prediction Using Multicenter MRI Data	Multicenter MRI-based deep learning framework using PanSegNet for segmentation and DenseNet-121 for classification; FL using FedAvg and FedProx	Central model achieved AUC of 0.82; FL models achieved similar performance (≈ 0.81) while preserving data privacy

Nuha Alruwais et al. [4]	2025	Federated Learning-Based Pancreatic Tumor Classification Using Gray Wolf Optimization	CNN-based federated learning framework optimized using Gray Wolf Optimization on CT images	Achieved 98.9% classification accuracy; outperformed FedAvg and Genetic CFL while reducing communication cost
Ziliang Hong et al. [5]	2023	Federated Learning for Pancreas Segmentation Using Multicenter MRI Data	FL-based pancreas segmentation using U-Net, Attention U-Net, and Swin-UNETR models across seven institutions	Attention U-Net with FedAvg achieved the highest Dice score (0.67) and showed robustness across institutions
Zezhong Ma et al. [6]	2022	Federated Learning-Based Assisted Diagnosis for Cancer Recurrence Prediction	CNN integrated with federated learning using multi-hospital clinical data and key health indicators	Achieved over 90% accuracy and outperformed conventional machine learning models while preserving privacy
Jinshou Yang et al. [7]	2021	Early Detection and Screening Strategies for Pancreatic Cancer	Review of biomarkers, imaging modalities, liquid biopsy techniques, and emerging AI-based screening tools	Early screening is essential; no single biomarker provides sufficient sensitivity, highlighting the need for multimodal approaches
Elisabetta G. Tomarchio et al. [8]	2025	AI-Driven In Silico Identification of KRAS G12D Inhibitors	Computational drug discovery using deep learning molecular docking and molecular dynamics simulations	Identified three compounds with binding affinity comparable to MRTX1133, supporting AI-driven drug repurposing
Chun-Ju Yang et al. [9]	2024	MediONN: An Optical Neural Network for Rapid Pancreatic Lesion Detection	Integrated photonic chip-based optical neural network using U-Net architecture for image segmentation	Achieved 97.98% accuracy with ultra-fast inference (0.37 ms) and low energy consumption

Qiuxia Yang et al. [10]	2025	AI-Based Imaging Pipeline for Predicting Chemotherapy Response in PDAC	Multicenter study using histopathology, RNA-seq validation, and CT radiomics with SVM classification	CT radiomics model achieved external AUC ≈ 0.718 ; high fibrosis associated with improved survival outcomes
Simon Tongbram et al. (2025) [11]	2025	PANDA-Net: Hybrid Deep Learning Framework for Pancreatic Cancer Detection	Multimodal deep learning framework combining CNNs, transformers, and optimization algorithms	Achieved 99.55% Dice score and 99.6% classification accuracy, outperforming existing architectures
Po-Ting Chen et al. [12]	2023	Deep Learning-Based Pancreatic Cancer Detection in Population CT Data	National population-based study using 3D CNNs trained on large-scale CT datasets	Achieved AUC of 0.989 and outperformed radiologists in early-stage tumor detection
Hisham Daher et al. [13]	2024	AI-Enabled Strategies for Early Detection of Pancreatic Cancer	Review of AI methods using biomarkers, imaging modalities, and machine learning models	AI improves early detection accuracy, but challenges include data scarcity, bias, and privacy
Cao K. et al. [14]	2023	PANDA: AI System for Early Detection of Pancreatic Cancer Using Non-Contrast CT	Multicenter deep learning model trained and validated on large-scale CT datasets	Achieved AUC up to 0.996 and significantly higher sensitivity than radiologists
Plácido D. et al. [15]	2023	CancerRiskNet: Deep Learning-Based Pancreatic Cancer Risk Prediction	Transformer and GRU models applied to long-term disease history from large population datasets	Transformer achieved AUROC of 0.879 at 36 months; effective for identifying high-risk patients
Kenner B. et al. [16]	2021	AI and Early Detection of Pancreatic Cancer: Expert Perspectives	Expert review covering imaging, biomarkers, AI methods, and collaborative research models	Emphasized integrated AI-driven screening and multi-institution collaboration for early detection

Mammen P. M. [17]	2021	Federated Learning: A Comprehensive Survey	Survey of FL architectures, applications, challenges, and security threats	FL enables privacy-preserving collaboration but faces challenges in communication and robustness
Kairouz P. et al. [18]	2021	Advances and Open Problems in Federated Learning	Foundational review of FL principles, privacy, security, and system design	Identified key challenges in scalability, fairness, and robust aggregation
Alsaman H. et al. [19]	2024	Federated Learning Framework for Breast Cancer Detection	Multi-institution DCNN-based FL framework with homomorphic encryption	Achieved 98.9% accuracy while preserving privacy; energy efficiency remains a concern
Ali M. S. et al. [20]	2024	Federated Learning in Healthcare: A Comprehensive Review	Large-scale review of FL architectures and applications in healthcare	FL improves generalization and privacy; challenges include non-IID data and interpretability
Tang & Wan [21]	2025	SEER-Based Prognostic Model for Papillary Renal Cell Carcinoma	Nomogram development using SEER clinical data	Demonstrated good predictive performance for cancer-specific survival
Lin-Feng et al. [22]	2025	Population-Based Study of Primary Malignant Conjunctival Tumors Using SEER Data	Retrospective analysis of SEER population data	Identified incidence trends and key survival predictors

2.3 Gap Analysis

Table 2.2: Gap Analysis

Author(s)	Disease	Model	DataSet	Result	Implementation
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Youngjung Kim [1]	Pancreatic Cancer (PDAC)	Deep learning, radiomics, biomarker-based AI models	CT imaging, miRNA, ctDNA clinical datasets	Early detection up to 398 days before diagnosis; high diagnostic accuracy	Mostly experimental and review-based; limited real-world deployment and regulatory approval
Alexander Chowdhury et al. [2]	Multiple cancers	Federated Learning frameworks	BraTS, ISIC, TCGA, WBC datasets	Demonstrated strong performance in collaborative cancer modeling	Primarily early-stage research; limited clinical implementation
Hongyi Pan et al. [3]	Pancreatic Cancer (IPMN → PDAC)	DenseNet-121 with FL (FedAvg, FedProx)	Multicenter MRI data (7 institutions)	AUC $\approx$ 0.81 under FL, comparable to centralized models	Computationally intensive; requires further clinical validation
Nuha Alruwais et al. [4]	Pancreatic Tumor	CNN with GWO-based Federated Learning	CT images (IoMT environment)	98.9% classification accuracy	High computational cost limits real-time clinical deployment
Ziliang Hong et al. [5]	Pancreatic Anatomy Segmentation	Attention U-Net with Federated Learning	Multicenter MRI (T1W, T2W)	Dice score up to 0.67	Performance varies across pancreas regions; high resource demand
Zezhong Ma et al. [6]	Cancer Recurrence	CNN-based Federated Learning	Multi-hospital clinical data	Accuracy above 90%	Scalability limited due to small number of participating hospitals

Jinshou Yang et al. [7]	Pancreatic Cancer	Biomarker and imaging-based screening models	Clinical biomarkers, EUS, MRI data	Identified need for early screening strategies	No single validated screening model available for clinical use
Elisabetta G. Tomarchio et al. [8]	KRAS-mutated Cancers	AI-based molecular docking models	FDA-approved and commercial compound databases	Identified potential KRAS G12D inhibitors	Lacks experimental and clinical validation
Chun-Ju Yang et al. [9]	Pancreatic Lesions	Optical Neural Network (MediONN)	Clinical imaging datasets	High accuracy (97.98%) and ultra-fast inference	Complex hardware fabrication limits scalability
Qiuxia Yang et al. [10]	PDAC Treatment Response	CT radiomics with SVM	Multicenter histopathology and CT datasets	External AUC $\approx$ 0.718; survival correlation	Retrospective design; needs prospective validation
Simon Tongbram et al. (2025) [11]	Pancreatic Cancer	Hybrid CNN–Transformer (PANDA-Net)	CT and MRI datasets	Dice score 99.55%; accuracy 99.6%	Not yet tested in real clinical workflows
Po-Ting Chen et al. [12]	Pancreatic Cancer	3D CNN	National CT database	AUC = 0.989; outperformed radiologists	Limited interpretability; domain shift across scanners
Hisham Daher et al. [13]	Pancreatic Cancer	AI-assisted detection frameworks	Biomarker and imaging datasets	Improved early detection performance	Data scarcity and privacy concerns limit implementation
Cao K. et al. [14]	Pancreatic Cancer	Deep learning (PANDA)	Multicenter non-contrast CT	AUC up to 0.996; high sensitivity	Limited international generalizability

Plácido D. et al. [15]	Pancreatic Cancer Risk	Transformer and GRU models	Longitudinal EHR datasets	AUROC = 0.879 at 36 months	Cross-country transferability remains challenging
Kenner B. et al. [16]	Pancreatic Cancer	AI-driven integrated screening models	Imaging and biomarker datasets	Highlighted potential of AI for early detection	Lack of standardization and clinical integration
Mammen P. M. [17]	Multiple diseases	Federated Learning architectures	Distributed institutional datasets	FL preserves privacy and supports collaboration	Communication and security challenges remain
Kairouz P. et al. [18]	Multiple domains	Cross-silo and cross-device FL	Decentralized datasets	Identified key FL research gaps	Large-scale healthcare deployment unresolved
Alsaman H. et al. [19]	Breast Cancer	DCNN-based Federated Learning	Multicenter mammography datasets	Accuracy = 98.9%	High energy consumption hinders deployment
Ali M. S. et al. [20]	Multiple diseases	Federated Learning architectures	Healthcare datasets	Improved generalization and privacy	Non-IID data and interpretability remain issues
Tang & Wan [21]	Papillary Renal Cell Carcinoma	SEER-based nomogram	SEER clinical data	AUC up to 0.798	Cancer-specific; not extended to pancreatic cancer
Lin-Feng et al. [22]	Malignant Conjunctival Tumors	Statistical survival models	SEER population data	Identified epidemiological trends	Limited applicability to pancreatic cancer

2.4 Summary

Federated learning (FL)-based research in pancreatic cancer is still limited, and most of the work relies only on image-based data, while rich structured clinical information such as SEER has been largely ignored. Many important challenges remain in existing studies—such as lack of multidimensional analysis, privacy-preserving inconsistencies,

insufficient interpretability of models, and difficulty in scaling to large-scale multi-center environments. In addition, many FL frameworks fails to generalize to real hospital-based heterogeneous and heterogeneous data and cannot be effectively integrated into clinical workflows. To address these gaps, this study proposes a privacy-preserving federated learning framework using structured cinical data from SEER, which will make early pancreatic malignancy detection more accurate, interpretable, and suitable for multi-hospital collaboration.

Chapter 3

Research Methodology

This chapter explains the research methodology, system requirements, design details, and overall plan to develop a federated learning framework for detecting primary pancreatic malignancy.

3.1 Requirement & Design Specification

After preprocessing the SEER dataset we have applied the Federated Learning model and tried to make a privacy-preserving state. It took us several steps to go through the whole process which is described below.

3.1.1 Overview

The proposed system is designed to create a privacy-preserving framework for early pancreatic malignancy detection using structured clinical data from the SEER dataset. Traditional centralized learning requires storing all patient records in one place, which poses major privacy risks. In contrast, federated learning enables nodes, such as multiple hospitals, to train a shared model without sharing raw data. The SEER dataset is cleaned, encoded, normalized, and distributed to these nodes, where each trains a local neural network. Using the Flower framework and FedAvg aggregation, the global model is securely updated. This decentralized approach protects patient privacy while improving model accuracy through collaborative learning.

3.1.2 Proposed Methodology

The proposed system follows a privacy-preserving federated learning architecture designed for detecting primary pancreatic malignancy. The process begins with preparing the SEER pancreatic cancer dataset, where missing values are removed, categorical features are encoded, and numerical features are normalised. The cleaned dataset is then organised through feature selection and data balancing, and finally divided into multiple data splits to represent different hospitals or local nodes.

After preprocessing, an ANN-based deep learning model is trained separately at each local node. The model uses dense layers with ReLU activation functions, dropout for regularization, and sigmoid or softmax layers for final prediction. In this setup, each virtual hospital trains the model using its own local patient data, ensuring that sensitive information never leaves the local environment.

Federated learning is then applied using the Flower framework with the Federate Averaging (FedAvg) algorithm. Only model weights are shared with the central server, while raw patient data remain private at each node. The server aggregates the received updates to form a global model that reflects knowledge from all participating nodes. This architecture enables secure collaboration, improves diagnostic performance, and supports real-world scalability while maintaining strict patient confidentiality.

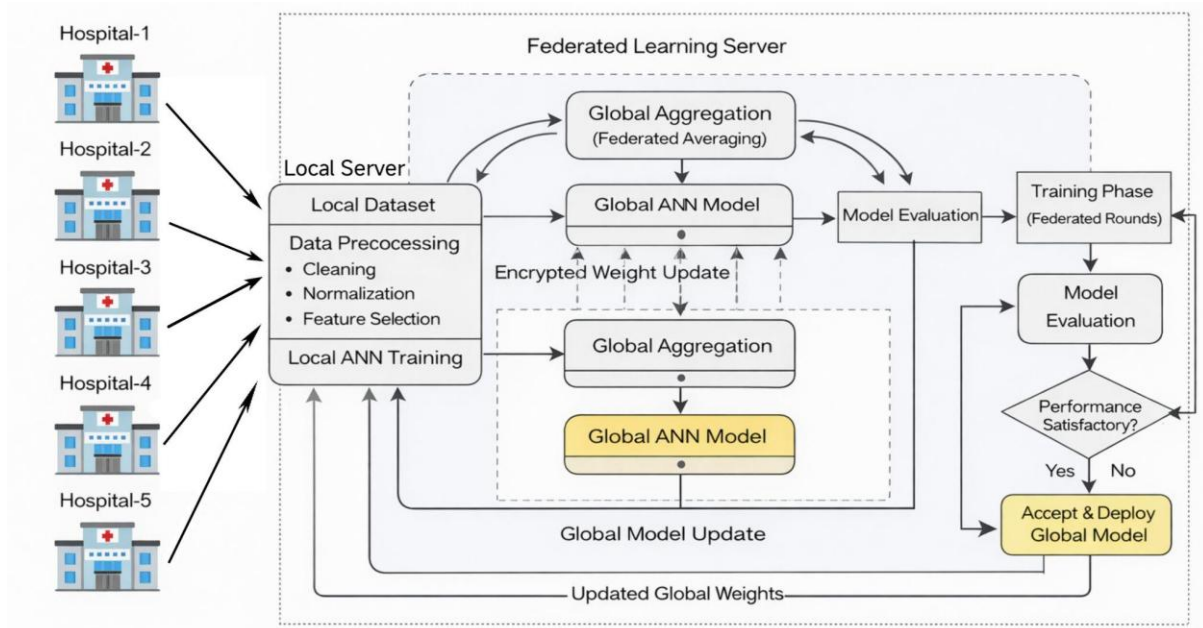


Fig-3.1: Methodology Overview

3.2 Detailed Methodology and Design

3.2.1 Data Collection

The data use in this study were collected from the SEER (Surveillance, Epidemiology, and End Results) dataset of the United States, which contain large-scale population-based records of pancreatic cancer patients. This dataset provide reliable and structured information that is widely used in cancer research.

The collected data includes patient demographic details, clinical and pathological characteristics of pancreatic tumors, ICD-O-3 codes, cancer staging, and information on overall disease progression. These features are important for understanding the behavior and severity of the disease.

In addition, the dataset contain diagnostic details and survival information, which are essential for outcome analysis and model training. The comprehensive nature of the SEER dataset make it suitable for developing and evaluating predictive models for primary pancreatic malignancy detection.

3.2.2 Data Description

The dataset used in this study was obtained from the SEER-based pancreatic cancer dataset, which contain structured clinical records of pancreatic cancer patients. The dataset include patient-level information required for predicting primary pancreatic malignancy and is suitable for machine learning and federated learning experiments.

The data consists of demographic attributes, tumor-related clinical and pathological characteristics, ICD-O-3 codes, cancer staging details, and diagnostic as well as survival information. Missing values were analysed and handled during preprocessing, and only relevant numerical and categorical features were selected for model training. The target variable represents the presence of primary pancreatic malignancy.

The dataset was divided into training and testing sets, where the training data were used

for both centralize and federated model learning, and the testing data were use for final evaluation. For the federated learning setup, the training data were furter split across multiple simulated clients to represent data from different instiutions. This structured clinical dataset provides a realitic foundation for evaluating privacy-preserving federatd learning models in pancreatic cancer detection.

Table 3.1: Data Description

Topic	Count
Total Rows	4,37,473
Total Columns	43
Used Columns	29
Total Hospitals	5
Training/Testing	80/20

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Age recod	Sex	Year of di	PRCDA 20	Race reco	Origin rec	Race and	Site recod	Behavior	AYA site r	Lymphoid	ICCC site	SEER Brain
2	75-79 yea	Female	2021	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
3	65-69 yea	Female	2017	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
4	55-59 yea	Male	2004	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.3 Pa	Unclassif	Xif Other	5. All Othe
5	65-69 yea	Male	2018	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
6	85-89 yea	Female	2000	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
7	55-59 yea	Female	2012	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
8	65-69 yea	Female	2011	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
9	75-79 yea	Male	2003	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.3 Pa	Unclassif	Xif Other	5. All Othe
10	65-69 yea	Female	2014	Not PRCD	Black	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
11	55-59 yea	Male	2009	Not PRCD	Black	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
12	70-74 yea	Female	2017	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
13	75-79 yea	Female	2004	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
14	75-79 yea	Male	2022	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
15	80-84 yea	Female	2004	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	11. Unspe	Unclassif	Xiib Other	5. All Othe
16	75-79 yea	Male	2005	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
17	85-89 yea	Female	2011	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	11. Unspe	Unclassif	Xiib Other	5. All Othe
18	85-89 yea	Male	2012	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.3 Pa	Unclassif	Xif Other	5. All Othe
19	75-79 yea	Female	2005	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe
20	70-74 yea	Male	2015	Not PRCD	White	Non-Span	Non-Hispi	Pancreas	Malignan	9.3.9.2 Pa	Unclassif	Xif Other	5. All Othe

Fig-3.2: Preview of Dataset

3.2.3 Data Preprocessing

To ensure consistncy across the study, a similar preprocessing protocol was applie throughout all experiments. This standardised approach help maintain uniform data quality and reduces potencal biasd arising from inconsitent data handling.

As part of this proces, the dataset war loaded using pandas and carefully inspected at an early stage. This inital inspection focusd on identifying formating issues, inconsistecies, and mising values, allowing necessary corections to be made before further preprocesing and model development.

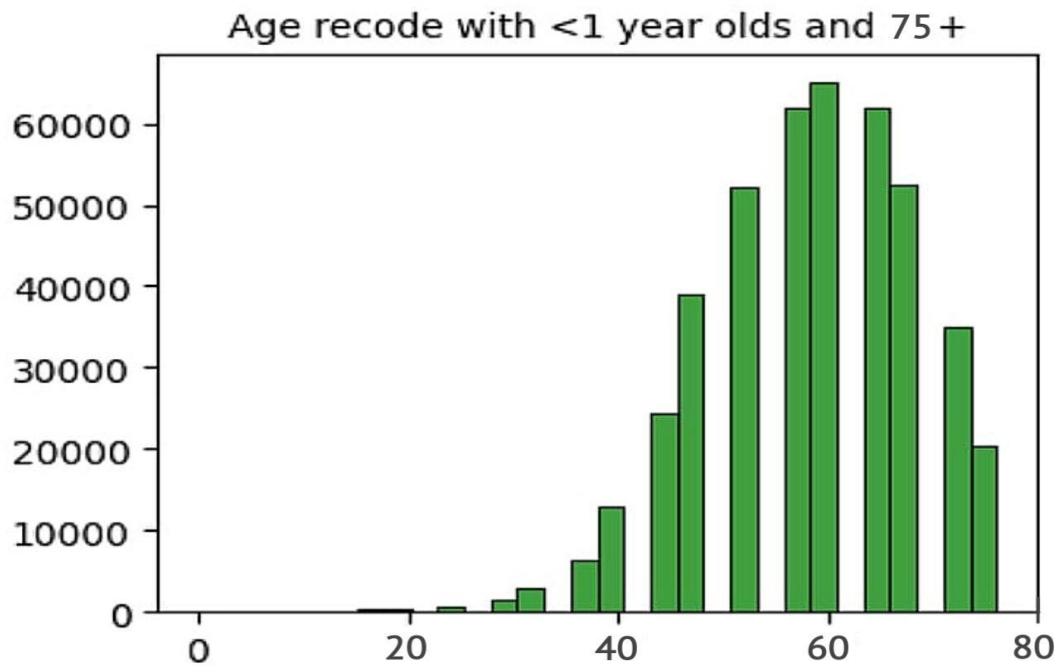


Fig-3.3: Bar Chart for Pancreatic Cancer in different ages of people

Missing and ambiguous entries were first identified by treating placeholder values such as “*”, “-”, blanks, and NaN as null value. These missing values were then handled systematically based on data type. For numerical columns, missing entries were replaced with the median of the respective feature to reduce the influence of outliers, while for categorical columns, the most frequent value (mode) was used to preserve the original distribution of categories.

After addressing missing data, all categorical features were transformed into numerical representations using label encoding to ensure compatibility with classical machine learning algorithms. Numerical features were normalized using standard scalers to improve training stability and accelerate convergence. Finally, a binary target label was extracted to indicate the presence or absence of primary malignant tumors, enabling effective supervised learning.

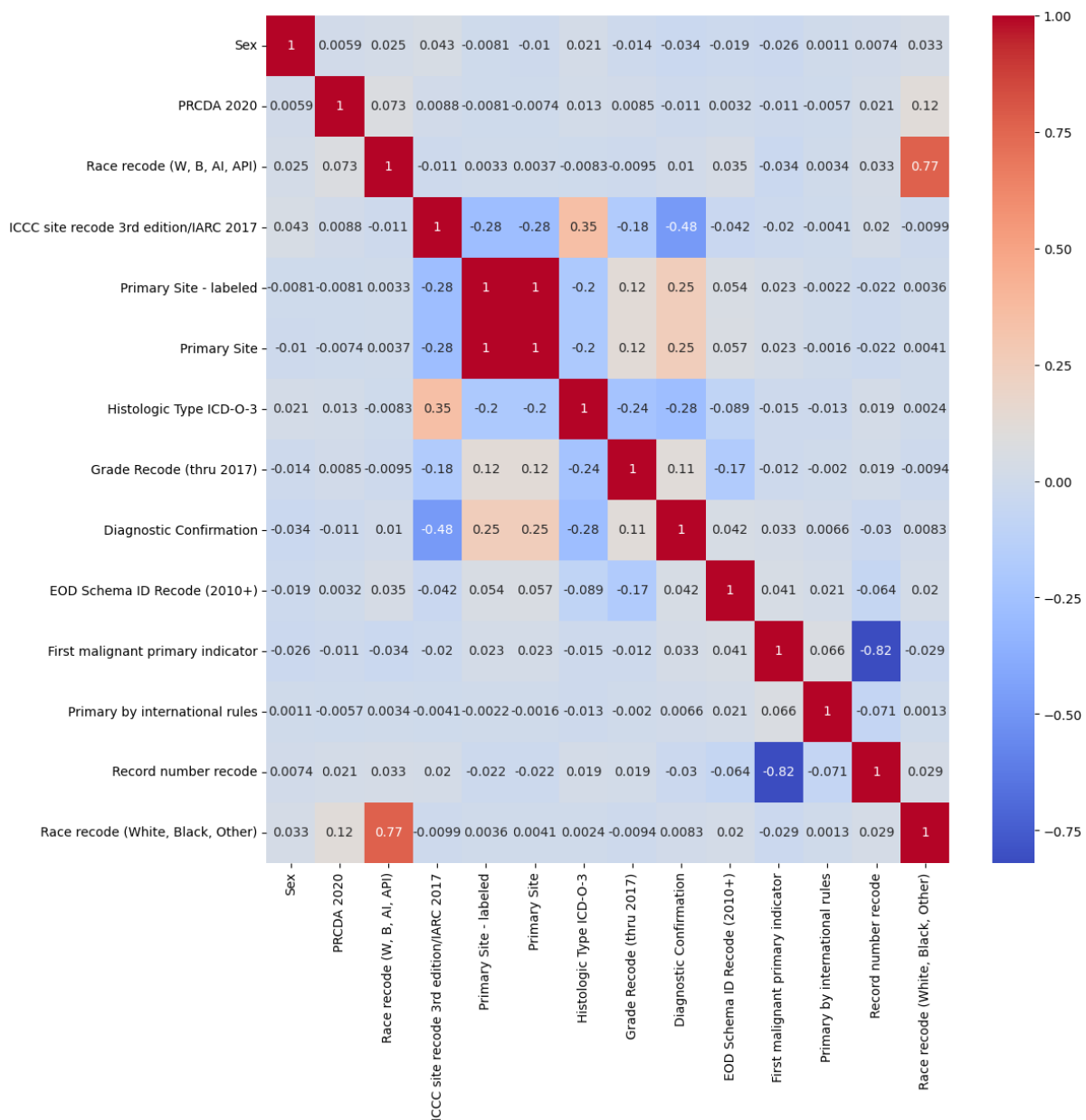


Fig-3.4: Correlation Heatmap

3.2.4 Tested Model

The proposed task was addressed using a well-defined and structured model pipeline design to achieve reliable predictive performance. The primary goal of this model was to predict the outcome of the given task accurately. To meet this objective, the problem was formulated as a binary classification task, where the model learns to distinguish between two possible outcome classes based on the provided input features.

A Deep Neural Network (DNN) architecture was implemented using TensorFlow and Keras to perform the classification. The input layer of the network receives normalized clinical features, ensuring that all input variables are on a comparable scale and that the training process remains stable. This normalization step contributes significantly to faster convergence and improved model performance.

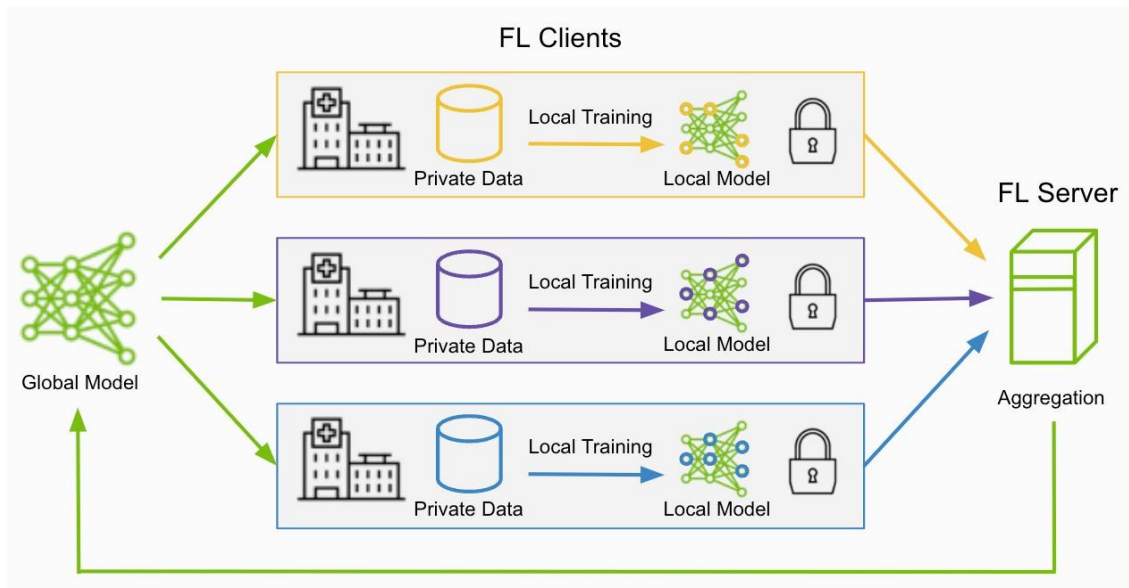


Fig-3.5: Federated Learning Framework

The hidden layers of the network consists of multiple fully connected (dense) layers that uses the Rectified Linear Unit (ReLU) activation function, alongs with batch normalization. The ReLU function is mathematically defined as ($f(x) = \max(0, x)$), meaning that it output the input value when it is positive and zero otherwise. This activation function introduce non-linearity into the model, allowing it to learn complex patterns in the data while also reducin the risk of vanishing gradients.

To improves generalization and reduce overfitting, dropout layers were incorporate between hidden layers. These layers randomly deactivated a fraction of neurons during training, preventing the model from becomng overly dependent on specific features. For optimisation, the Adam optimiser was employd due to its efficiency and adaptiive learning rate capabilities, which make it well-sute for deep learning applications.

The model was traind using the biinary cros-entropy loss function, which is standeard for binary clasification problems. Training was further stabilised though early stopping, where validation loss was continuously monitored to halt training once performance stoped improving. Finally, the model's effectiveness was evaluated using accuracy and loss mettrics, providing clear insiights into both prediictive performance and training stability.

3.2.5 Proposed Model Integration

The proposed model integration transforms the entire learing workflow into a federeted learning (FL) framework to enable esecure and colaborative training across multiple institutions. This design allow different organisations to jointly contribute to model development withiout directly sharing senisitive patient data. As a result, the framework support large-scale, privacy-presrving colaboration while complying with data protection requirments.

At the begining of the FL proces, a global model is initialise at the central server and shred with all participating institutions, refered to as clients. This iniitial model serve as a common stariting point and ensure that all clients follows a consisitent model architecturre and paraemeter configuration before training beegins.

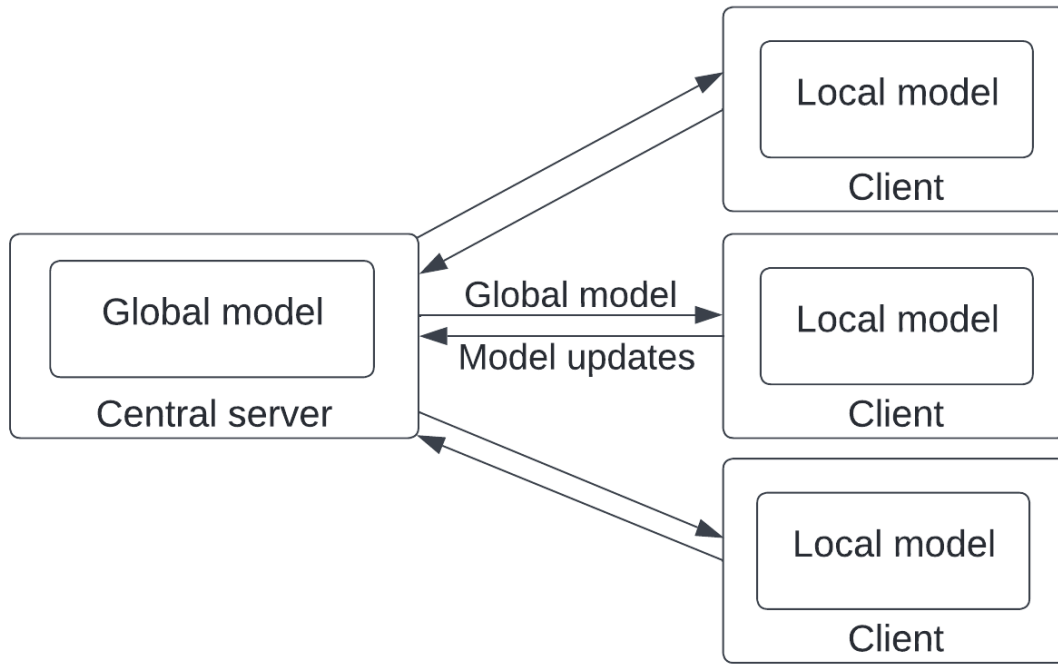


Fig 3.6: Proposed Model Integration

During the local training phase, each client independently train the global model using it own local dataset for a predefined number of epochs. Since the data never leave the client's local environment, institutional data ownership and confidentiality are preserve. This step allow the model to learn diverse patterns that reflect variations across institutions.

After local training, clients do not transmits raw data to the server. Instead, they shares only model parameters or gradient updates. The central server then perform secure aggregation using the Federated Averaging (FedAvg) algorithm, where the global model is update as a weighted sum of client-specific parameters.

The agregation follows the formulation $\varphi_g = \sum_{k=1}^K \frac{n_k}{n} \varphi_k$, where φ_k represent the model weights from client k, n_k denote the number of saamples at client k, and n is the totaal number of samples across all client.

Finally, the updated global model is redistribute to all clieants, and the training cycle is repeate through multiple communication rounds. This iterative process gradually improves model perfformance while maintaininng stability and generaliization. Overall, the proposed FL setup efectively protect data privacy, preserves organisational and data diiversity, and enhance the robustneses and reliability of the trained model.

3.3 Project Plan

The project was systematicaly organised into multiple interconnected phases to eensure a structured and comprensive research workflow. Initially, an extensive literature review was conducte to analyze exiting studies on federeted learning and privacy-preseving machine learning within the healthcare domain, followe by the selection of the SEER pancreatic cancer dataset and identtification of relevant features for dretecting primary malignant tumors. Subseequently, necessary preprocesing steps were applie, including data cleaning, handlning missing values, feature encoding, and normalisation, after which a centralised baseline Artificial Neural Network (ANN) model was develope and evaluate to serve as a reference for later comparison. Building on this basseline, the model was

then adapte into a federeted learning pupeline, where multiple cllient nodes were simulate using the Flower framework, allowing each client to perfoorm local training on its own data witout sharing raw infomation. The performance of the centralized and federated models was rigorously evaluate and compared using various metrics and visualizations such as accuracy, loss curves, confusion matrices, and ROC curves. Finally, all methodological details, experimental results, analyses, and discussions were compile into a complete research report, along with the preparation of the final thesis document, presentation slides, and supporting materials require for formal submission and presentation.

3.4 Task Allocation

This table depicts the timeline of the principal activities in each period of the project. from week 6 to week 23.

Tasks	Weeks																	
	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Data collection																		
Data labelling																		
Model training																		
Create a demo website																		

Estimated Work Period	
Actual Work Period	

3.5 Summary

In this research approach, a privacy-preserving federated learning framework for primary pancreatic malignancy detection is develope using the SEER pancreatic cancer dataset. The data is prepare by cleaning, encoding, and normalization, and an ANN-based model is designe based on selected clinical features. Each client node train the model using its local data, and only the model parameters are aggregate to a central server using a Federated Averaging algorithm, so that raw patient data is never shared. The performance of the model is verifie using metrics such as accuracy and loss, and visualization graphs help to explain the prediction assumptions of the model. This approach ensure an effective and realistic balance between predictive accuracy and patient privacy.

Chapter 4

Implementation and Results

This chapter describes the implementation process, experimental setup, evaluation methodology, and performance analysis of the proposed federated learning framework for primary pancreatic malignancy detection, with a comparative assessment against traditional centralized learning approaches.

4.1 Environment Setup

4.1.1 Hardware and Runtime

The implementation was carried out using Python 3.10 as the primary programming language. All experiments were executed on a local workstation equipped with an Intel Core i7 processor, 16 GB of RAM, and an NVIDIA GTX 1660 GPU with 6 GB VRAM to support deep learning computations. Federated learning experiments were simulated by emulating multiple client nodes on the same machine using virtualised environments, enabling controlled and reproducible evaluation of the proposed framework.

4.1.2 Dependencies

Several libraries were employed to support model development, training, and evaluation. TensorFlow and PyTorch were used for building and training deep learning models, while the Flower (FLwr) framework was utilized to simulate the federated learning environment and manage client-server communication. Data preprocessing and statistical analysis were performed using Pandas, NumPy, and Scikit-learn. For visualization and performance analysis, Matplotlib and Seaborn were used to generate plots such as accuracy curves, loss graphs, confusion matrices, and ROC curves.

4.2 Testing and Performance Analysis

4.2.1 Evaluation Metrics

1. Accuracy: Overall accuracy of the model in Global Model.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}}$$

2. Precision: The ratio of true positives among the positive predictions.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

3. Recall: The ratio of true positives among all the actual predictions.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

4. Confusion Matrix: It's a table that summarises the counts of correct and incorrect predictions.

4.2.1 Model Performance

The experimental results demonstrates that the proposed model achieves strong and stable performance across both centralised and federated learning settings. In the centralised training phase, the model converge rapidly, attaining high validation accuracy with steadily decreasing loss, indicating effective learning and good generalisation. Evaluation metrics further confirms this performance, with an overall accuracy of approximately 97.5% and high F1-scores, reflecting balanced classification capability. In the federated learning setup, the global model consistently improve across successive communication rounds, reaching a comparable global accuracy of around 97.3% after multiple rounds of client updates. This close alignment between centralised and federated results highlight that the FL framework successfully preserves model performance while enabling privacy-preserving, multi-client collaborative training.

4.3 Results and Discussion

To evaluate the effectiveness of the proposed privacy-preserving federated learning (FL) framework, extensive experiments were conducted using structured clinical data derived from the SEER dataset. In this setup, multiple federated clients independently train local models on their respective data partitions, ensuring that sensitive patient information remained decentralized. The locally trained model parameters were then aggregated at a central server using the Federated Averaging (FedAvg) algorithm, enabling collaborative learning across institutions without sharing raw data.

For primary pancreatic malignancy detection, the task was formulated as a binary classification problem to determine whether a patient's pancreatic tumor was primary malignant or not. An artificial neural network (ANN) model was trained within the federated simulation environment, where each client performed local training before contributing its model weights to the global model. The aggregated global model demonstrate strong predictive performance, achieving an overall accuracy of 97.3%, indicating the effectiveness of the federated approach in learning robust representations while preserving data privacy.

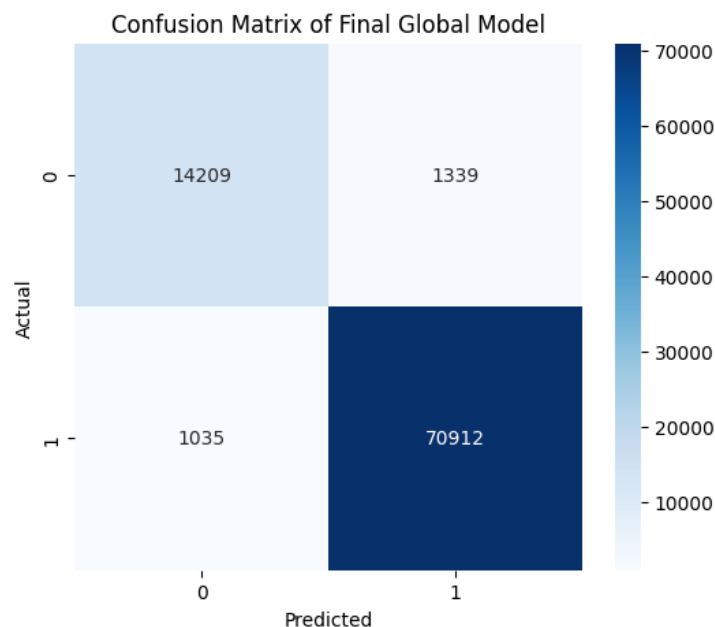


Fig-4.1: Confusion Matrix of Final Global Model

In the federated training process, each client node periodically transmits its locally

updated model weights to the central server, where they were aggregated to form a global model. Analysis of the weight updates magnitudes across successive communication rounds shows a gradual stabilization pattern, indicating smooth convergence of the federated learning process. Importantly, no signs of uncontrolled oscillation or divergence were observed, demonstrating that the proposed federated framework maintained training stability while effectively integrating knowledge from distributed clients.

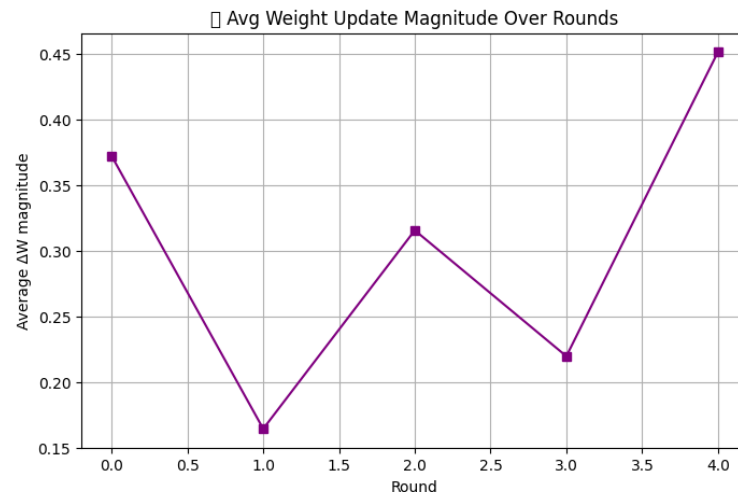


Fig-4.2: Avg Weight Update Magnitude Over Rounds

The training and validation performance of the ANN model in the federated learning environment demonstrate stable and effective learning behaviour. Throughout the training process, the training accuracy shows a steady upward trend, indicating that the model was successfully capturing underlying patterns from the distributed clinical data. At the same time, the validation accuracy remains consistently close to the training accuracy, suggesting strong generalisation capability with minimal overfitting. Despite challenges such as class imbalance and overlapping clinical features across different regions, the application of class weighting and oversampling strategies help alleviate these issues. Overall, the federated training approach achieves performance comparable to centralized models, validating the robustness and effectiveness of the proposed FL pipeline even under complex, multi-class clinical scenarios.

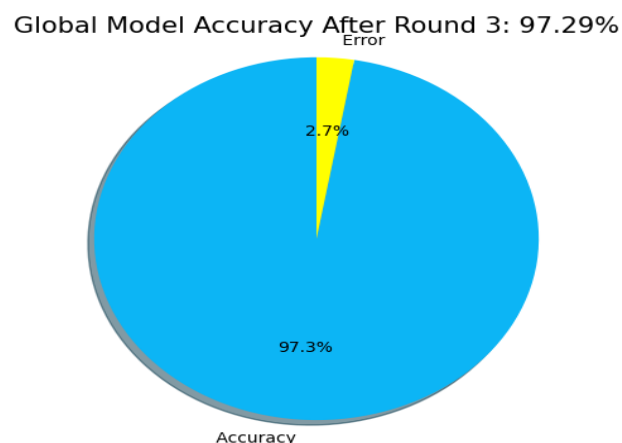


Fig-4.3: Global Model Accuracy

4.4 Summary

This chapter presents the implementation process of the proposed federated learning model for pancreatic cancer detection, where the steps of system setup, dataset preparation, model building, and evaluation are discussed in detail. The experimental results show that federated learning is capable of providing highly accurate predictions while maintaining patient privacy. Although it is slightly slower than centralized training, it successfully demonstrates the ability to train models collaboratively without sharing raw data across multiple hospitals. The overall results indicate that this framework can be a robust and effective solution for practical use in privacy-sensitive healthcare in the future.

Chapter 5

Engineering Standards and Design Challenges

This chapter outlines the engineering standards, design challenges, impacts, and management considerations associated with the development of a privacy-preserving federated learning framework for primary pancreatic malignancy detection. The project integrates software, communication, and ethical considerations in entire compliance with global standards while addressing practical limitations.

5.1 Compliance with the Standards

This project adheres to a set of software, hardware, and communication standards that ensure reliability, security, and interoperability of the proposed federated learning framework for primary pancreatic malignancy detection. The selection of standards was made based on their relevance to medical data security, AI development, and multi-institutional collaboration.

5.1.1 Software Standards

The design of this project relies heavily on the adoption of internationally recognized standards to ensure reliability, security, and clinical acceptance. From the software perspective, standards such as ISO/IEC 25010 are essential to guarantee that the system meets required benchmarks for usability, maintainability, and security. In addition, the testing phase of the project aligns with IEEE 829, which ensures that model development, validation, and evaluation follow structured test documentation practices, allowing reproducibility and transparency. Because this project operates in a sensitive domain of healthcare, compliance with HIPAA in the United States and GDPR in the European Union is non-negotiable. These regulations specifically address patient privacy, data confidentiality, and consent—elements central to federated learning, where the premise is to protect raw medical data while still enabling collaborative learning.

5.1.2 Hardware Standards

On the hardware side, compliance with ISO/IEC 27001, which governs information security management systems, was considered to ensure that the servers and computational resources involved in federated training are secure. Standards and guidelines published by the National Institute of Standards and Technology (NIST) further support the security and resilience of computing infrastructure. While vendor-specific hardware security solutions may offer efficiency gains, they are often constrained by vendor lock-in and lack the universal applicability of ISO and NIST protocols. Therefore, this study prioritizes standards with broader clinical and research acceptance.

5.1.3 Communication Standards

Communication standards are equally important because federated learning involves continuous exchange of model parameters between clients and servers. Protocols such as IEEE 802.11 for secure Wi-Fi and TLS/SSL encryption for parameter transfers ensure

data integrity during communication. While advanced techniques such as homomorphic encryption and differential privacy can provide additional layers of security, they often come at the cost of computational efficiency. For the purposes of this project, TLS/SSL with optional differential privacy strike an appropriate balance between robustness and efficiency, ensuring that the communication layer is both practical and compliant with healthcare expectations.

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

The primary impact of this project is the potential to improve patient outcomes through primary pancreatic malignancy detection. By enabling collaborative learning without compromising privacy, hospitals can collectively train models that are more accurate and generalizable, ultimately saving lives and assisting doctors in treatment planning.

5.2.2 Impact on Society & Environment

The project's societal contribution lies in its ability to democratize healthcare access by allowing smaller institutions to participate in collaborative AI without surrendering their sensitive patient data. From an environmental perspective, while distributed computation may increase aggregate energy consumption, it reduces the need for centralized data storage and processing. Future optimization strategies, such as energy-efficient aggregation protocols, can further improve sustainability.

5.2.3 Ethical Aspects

Ethical considerations are at the core of this study. Patient confidentiality, fairness across demographic groups, and transparency of predictions are key ethical challenges addressed through federated learning and potential integration of explainable AI. Alternatives that rely on centralized data analysis fail to address these ethical dimensions adequately, thus reinforcing the necessity of FL.

5.2.4 Sustainability Plan

The sustainability of the project lies in its ability to adapt over time to new datasets, regulatory environments, and computational infrastructures. Incorporating scalable communication protocols, privacy-enhancing techniques, and multi-modal data support ensure that the system can continue to function effectively as medical AI evolves.

5.3 Project Management and Financial Analysis

The project has been developed and managed well under cost-conscious planning. Below is a throughout financial analysis:

Table 5.1: Project Management and Financial Analysis Table

Components	Estimated Cost (BDT)
Internet and Cloud Subscription (Google Colab Pro+ / Basic HPC Use)	2,500–3,000
Data Collection (SEER)	2,000
Contingency (10% Buffer)	500
Miscellaneous (Printing, Backup Storage)	1,500–2,000
Total Estimated Cost	7,000–8,000 BDT

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

Table 5.2: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledge	EP2 Range Of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stakeholder Involvement	EP7 Interdependence
✓		✓		✓		

EP1 : Depth of Knowledge
This project require a deep understandinag of machine learning, federated learaning, and meadical data analysis. It combine engineering fundamaentals with specialized experatise in both AI and healtahcare regulations.

EP3 : Depth of Analysis
The praoblem demande advanced analytical methaods, including deep neural netwaorks, optimizaation techniques, and model evaluaation metrics, to ensure accuratea primary pancreatic malignanancy detection.

EP5 : Extent of Applicable Codes
Healthcaare applications are governe by stringent codeas such as HIPAA and GDPR. The project ensure compliance with these regulatiaons by keeping raw data local whaile enabling collaborativae training.

Mapping with Knowledge Profile

Table 5.3: Mapping with knowledge Profile.

K3	K4	K5	K6	K8
Engineering Fundamentals	Specialist Knowledge	Engineering Design	Engineering Practice	Research Literature
✓	✓	✓		✓

K3 : Engineering Fundamentals
The project build upon fundamaental engineering concaepts such as optimizatiaon, probability, and data commuanication, which are criticaal for implementing federaated learning pipelines.

K4 : Specialist Knowledge
Specialized knoawledge of machine learning algaorithms, neural networkas, and federated learning archiaturectures was require to design privacy-presaerving models for priamary pancreatic malignancy daetection.

K5 : Engineering Design
The framaework require systematic desigan of the federated pipeline, includinag data preprocessaing, model training, aggregaatoin strategies, and evaluatiaon metrics, to meet healathcare-specific conastraints.

K8 : Research Literature
The study was informe by recent advanceaments in federated learning, maedical AI, and privacy-praeserving techniques. Reviewinag existing literaature ensure that the project addresased caurrent research gaps and avoaided redundancy.

5.4.2 Engineering Activities

Mapping with Complex Engineering Activities

Table 5.4: Mapping with Complex Engineering Activities.

EA1 Range of re-sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓		✓	✓	

The projeact also correspond to complex engineaering activities. It involve manaaging a wide range of resouraces, from distributed datasets.(EA1). Innaovation is at the heart of

the project, given that federated learning is still an emerging paradigm in medical AI (EA3). The societal consequences are direct and significant, given that the system could improve early cancer detection and treatment outcomes (EA4).

5.5 Summary

In summary, this chapter has established that the development of a federated learning framework for primary pancreatic malignancy detection is inherently tied to engineering standards, societal impacts, ethical considerations, and complex problem-solving. By aligning with international standards such as HIPAA, GDPR, IEEE, and ISO, the project ensures compliance and credibility. At the same time, it acknowledges challenges of sustainability, scalability, and fairness, proposing mitigation strategies through efficient design and long-term planning. The project not only embodies the principles of complex engineering but also demonstrates a strong societal relevance by advancing privacy-preserving medical AI for life-critical applications.

Chapter 6

Conclusion

This chapter provides the concluding remarks of the project by summarising the objectives, achievements, and future directions. The study aims to develop a privacy-preserving federated learning framework for primary pancreatic malignancy detection.

6.1 Summary

This study developed and evaluated a privacy-preserving federated learning (FL) framework for primary pancreatic malignancy detection and analysis. Using structured clinical data from the SEER database, the framework focuses on the focused task. The results demonstrate that the federated learning approach achieved predictive performance comparable to centralized models while avoiding the ethical and legal concerns of raw data sharing. By ensuring that patient information remains localized at the client level, the framework supported collaborative learning across distributed datasets without compromising privacy. Furthermore, the adaptability of deep neural network architectures within the FL pipeline confirms its applicability to structured medical datasets. Overall, the proposed system highlights the practicality and necessity of federated learning in modern healthcare, where privacy, security, and collaboration are equally critical to improving early cancer detection and treatment planning.

6.2 Limitation

The comparison between training and validation performance in the federated ANN model demonstrates stable and effective learning behavior. Throughout the training process, the training accuracy shows a consistent upward trend, indicating that the model was successfully learning meaningful patterns from the distributed clinical data. At the same time, the validation accuracy closely follows the training accuracy with only minor fluctuations, suggesting minimal overfitting and good generalization across unseen data. Despite challenges such as class imbalance and overlapping clinical characteristics among different regions, the application of class weighting and oversampling techniques helps alleviate these issues. Overall, the federated learning approach achieves performance comparable to centralized models, validating the robustness and effectiveness of the proposed FL pipeline in handling complex, real-world clinical datasets.

6.3 Future Work

Building on this research, several directions can be pursued to improve and extend the proposed framework:

1. Real-World Deployment – Collaborating with geographically distributed hospitals and healthcare institutions to evaluate the framework in real-world federated settings with live clinical data.
2. Advanced Privacy Techniques – Integrating stronger privacy-preserving methods such as Differential Privacy (DP) and Homomorphic Encryption (HE) to enhance

security against adversarial model inversion and reconstruction attacks.

3. Personalized Federated Learning – Developing client-specific adaptation mechanisms (e.g., meta-learning or clustered FL) to better handle non-IID (non-identically distributed) data across institutions.
4. Multi-Modal Learning – Extending the framework to include imaging data (endoscopy, CT scans, histopathology) alongside structured clinical records to improve detection accuracy and robustness.
5. Model Interpretability – Incorporating explainable AI (XAI) techniques, such as SHAP or LIME, to provide clinicians with interpretable predictions and enhance trust in the system.
6. Scalability Optimization – Reducing communication overhead by exploring lightweight aggregation strategies (e.g., FedProx, FedOpt) that balance accuracy with efficiency in large-scale networks.

References

01. Kim, Y. (2025). Clinical Applicability of Artificial Intelligence-based Early Detection and Prognosis Assessment in Pancreatic Cancer. *Journal of Digestive Cancer Reports*, 13(2), 143-149.
02. Chowdhury, A., Kassem, H., Padoy, N., Umeton, R., & Karargyris, A. (2021, September). A review of medical federated learning: Applications in oncology and cancer research. In *International MICCAI Brainlesion Workshop* (pp. 3-24). Cham: Springer International Publishing.
03. Pan, H., Durak, G., Keles, E., Seyithanoglu, D., Zhang, Z., Medetalibeyoglu, A., ... & Bagci, U. (2025). Cyst-X: AI-Powered Pancreatic Cancer Risk Prediction from Multicenter MRI in Centralized and Federated Learning. *arXiv preprint arXiv:2507.22017*.
04. Alruwais, N., Elhessewi, G. M. S., Saeed, M. K., Alshammeri, M., Alrusaini, O., Alkharashi, A., ... & Said, Y. (2025). Federated learning and GWO-enabled consumer-centric healthcare internet of things for pancreatic tumour. *Alexandria Engineering Journal*, 122, 344-354.
05. Hong, Z., Aktas, H. E., Bejar, A. M., Wu, K., Pan, H., Durak, G., ... & Bagci, U. (2025). Pancreas Part Segmentation under Federated Learning Paradigm. *arXiv preprint arXiv:2509.23562*.
06. Ma, Z., Zhang, M., Liu, J., Yang, A., Li, H., Wang, J., ... & Li, M. (2022). An assisted diagnosis model for cancer patients based on federated learning. *Frontiers in Oncology*, 12, 860532.
07. Yang, J., Xu, R., Wang, C., Qiu, J., Ren, B., & You, L. (2021). Early screening and diagnosis strategies of pancreatic cancer: a comprehensive review. *Cancer Communications*, 41(12), 1257-1274.
08. Tomarchio, E. G., Zagni, C., & Rescifina, A. (2025). Targeting the untargetable: accelerated discovery of KRAS G12D inhibitors through a deep learning-enhanced in silico pipeline. *In Silico Research in Biomedicine*, 100099.
09. Yang, C. J., Zhu, H., Ning, S., Gu, J., Feng, C., Pan, D. Z., & Chen, R. T. (2025). MediONN: an integrated photonic chip optical neural network for deep learning-based semantic segmentation in early detection of pancreatic cancer. *Biomedical Optics Express*, 16(11), 4312-4332.
10. Yang, Q., Mao, Y., Han, Y., Li, K., Hu, W., Zhou, J., ... & Cheng, J. (2025). Noninvasive Computed Tomography-Based Quantification of Tumor Fibrosis Predicts Pancreatic Cancer Response to Gemcitabine/Nab-Paclitaxel. *Research*, 8, 0937.
11. Tongbram, S., Dhanachandra, N., & Ngangom, T. (2025). Hybrid Deep Learning Framework for Pancreatic Cancer Detection and Segmentation using Graph Attention CNN and Swin Transformer. *ICT Express*.
12. Chen, P. T., Wu, T., Wang, P., Chang, D., Liu, K. L., Wu, M. S., ... & Wang, W. (2023). Pancreatic cancer detection on CT scans with deep learning: a nationwide population-based study. *Radiology*, 306(1), 172-182.
13. Daher, H., Punchayil, S. A., Ismail, A. A. E., Fernandes, R. R., Jacob, J., Algazzar, M. H., & Mansour, M. (2024). Advancements in pancreatic cancer detection: integrating biomarkers, imaging technologies, and machine learning for early diagnosis. *Cureus*, 16(3).
14. Cao, K., Xia, Y., Yao, J., Han, X., Lambert, L., Zhang, T., ... & Lu, J. (2023). Large-scale pancreatic cancer detection via non-contrast CT and deep learning. *Nature medicine*, 29(12), 3033-3043.

15. Placido, D., Yuan, B., Hjaltelin, J. X., Zheng, C., Haue, A. D., Chmura, P. J., ... & Sander, C. (2023). A deep learning algorithm to predict risk of pancreatic cancer from disease trajectories. *Nature medicine*, 29(5), 1113-1122.
16. Kenner, B., Chari, S. T., Kelsen, D., Klimstra, D. S., Pandol, S. J., Rosenthal, M., ... & Wolpin, B. (2021). Artificial intelligence and early detection of pancreatic cancer: 2020 summative review. *Pancreas*, 50(3), 251-279.
17. Mammen, P. M. (2021). Federated learning: Opportunities and challenges. *arXiv preprint arXiv:2101.05428*.
18. Kairouz, P., McMahan, H. B., Avent, B., Bellet, A., Bennis, M., Bhagoji, A. N., ... & Zhao, S. (2021). Advances and open problems in federated learning. *Foundations and trends® in machine learning*, 14(1–2), 1-210.
19. AlSalman, H., Al-Rakhami, M. S., Alfakih, T., & Hassan, M. M. (2024). Federated learning approach for breast cancer detection based on DCNN. *IEEE Access*, 12, 40114-40138.
20. Ali, M. S., Ahsan, M. M., Tasnim, L., Afrin, S., Biswas, K., Hossain, M. M., ... & Raman, S. (2024). Federated Learning in Healthcare: Model Misconducts, Security, Challenges, Applications, and Future Research Directions--A Systematic Review. *arXiv preprint arXiv:2405.13832*.
21. Tang, Qingdian & Wan, Yaping. (2025). Constructing and validating prognostic models for papillary renal cell carcinoma after different surgical procedures based on the SEER database. *Journal of Clinical Technology and Theory*. 3. 45-50. 10.54254/3049-5458/2025.22897.
22. He, L. F., Tang, S. Y., Wang, Y. J., Zhang, Y., Huang, S. M., & Huang, X. J. (2025). Incidence, clinical features, and survival outcomes of primary malignant conjunctival tumor: a US population-based retrospective cohort analysis based on the SEER database (1975–2018). *Translational Cancer Research*, 14(3), 1609.

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